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Oefeningenreeks 5A nr. 6.5

$$\begin{aligned} & \int_0^9 e^{3x} dx \\ & t = 3x \left( dt = 3dx, dx = \frac{dt}{3} \right) \\ & \begin{cases} x = 0 \rightarrow t = 3 \cdot 0 = 0 \\ x = 9 \rightarrow t = 3 \cdot 9 = 27 \end{cases} \\ & = \int_0^{27} e^t \frac{dt}{3} \\ & = \frac{1}{3} \int_0^{27} e^t dt \\ & = \frac{1}{3} [e^t]_0^{27} \\ & = \frac{1}{3} (e^{27} - e^0) \\ & = \frac{1}{3} (e^{27} - 1) \end{aligned}$$

## References